

Binomial and Poisson Distributions

1. Bernoulli Distribution

Definition: A Bernoulli distribution is the simplest probability distribution that represents a random experiment with exactly two possible outcomes: *success* (usually denoted by 1) and *failure* (denoted by 0). Let p be the probability of success, then $1 - p = q$ is the probability of failure.

Probability Mass Function (PMF): $P(X = x) = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \end{cases}$

Mean and Variance: $\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p) = pq.$

Moment Generating Function (MGF): $M_X(t) = \mathbb{E}[e^{tX}] = (1 - p) + pe^t.$

Examples of Applicability:

- Tossing a coin once (Head = 1, Tail = 0).
- Whether a student passes an exam (Pass = 1, Fail = 0).
- Whether a machine works on a given day (Working = 1, Failure = 0).

2. Binomial Distribution

Definition: A binomial distribution gives the probability of obtaining a fixed number of successes in n independent Bernoulli trials, each with probability of success p .

PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

Mean and Variance:

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

Examples of Applicability:

- Number of heads in 10 tosses of a fair coin.
- Number of defective items in a batch of n when each item is independently defective with probability p .
- Number of patients recovering from a treatment, where each has independent probability p of recovery.

3. Poisson Distribution (as a Limiting Case of Binomial)

The Poisson distribution can be viewed as a limiting form of the Binomial distribution when

$$n \rightarrow \infty, \quad p \rightarrow 0, \quad \text{and} \quad \lambda = np \text{ is finite.}$$

In this case, the random variable X representing the number of successes in n independent Bernoulli trials follows approximately a Poisson distribution with parameter λ .

Definition: A Poisson distribution with parameter λ gives the probability of observing k events in a fixed interval when the events occur independently and with a constant average rate λ .

$$\lambda = np$$

is interpreted as the *expected number of occurrences* in the interval, where

- n = number of trials (large),
- p = probability of success in each trial (small), and
- λ = average number of successes over n trials (finite).

PMF:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

Mean and Variance:

$$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda.$$

4. Key Relationships

- The **Bernoulli** distribution is the basic building block for the **Binomial** distribution.
- The **Binomial** distribution represents the number of successes in n Bernoulli trials.
- The **Poisson** distribution can approximate the Binomial distribution when n is large and p is small, with $\lambda = np$.

Examples

1. A fair coin is tossed 5 times. Find the probability of getting exactly 3 heads.
2. A multiple-choice question has four options, only one of which is correct. If a student guesses the answer to 10 such questions, find the probability that he answers exactly 2 correctly.
3. A factory produces light bulbs, and each bulb has a 0.1 probability of being defective. Out of 12 bulbs selected at random, find the probability that exactly 2 bulbs are defective.
4. The probability that a person is left-handed is 0.12. In a group of 8 students chosen at random, find the probability that:
 - Exactly 2 students are left-handed.
 - At most 2 students are left-handed.
5. A computer server receives 20 independent access requests. Each request succeeds with probability 0.9. Find the probability that:
 - All requests succeed.
 - At least 18 requests succeed.
6. The probability that a machine part produced is defective is 0.02. A batch of 50 parts is inspected.
 - Find the probability that there are exactly 3 defective parts.
 - Using Poisson approximation, estimate this probability.
7. An e-mail system detects spam with a probability of 0.95. If $n = 15$ spam e-mails are sent, find:
 - The probability that all are correctly detected.
 - The probability that at least 13 are correctly detected.
8. The average number of students arriving late to a class is 2 per day. Using Poisson distribution, find the probability that on a given day:
 - No student arrives late.
 - Exactly 3 students arrive late.
9. A typing error occurs on average 1.5 times per page. Find the probability that randomly selected five pages contain no error.

10. On average, 3 customer complaints are received per hour at a help desk. Find the probability that in a particular hour:

- No complaints are received.
- More than 5 complaints are received.

Vehicles pass through a toll booth at an average rate of 5 per minute. Using Poisson distribution, find the probability that during a given minute:

- Exactly 4 vehicles pass.
- At most 3 vehicles pass.

11. Misprints in a book occur randomly with an average of 1 misprint per page. Find the probability that in 4 consecutive pages, there are altogether exactly 2 misprints.

12. Calls arrive at a call center according to a Poisson process at an average rate of 10 calls per hour. Find the probability that:

- No call arrives in the first 15 minutes.
- At least 3 calls arrive in 30 minutes.